

Flash ADC

StrongARM Latch Comparator and CMOS Encoder

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What is a Flash ADC?

- A Flash ADC is the **fastest ADC architecture**, as conversion happens in **one clock cycle**.
- Uses an array of comparators that compare input voltage with multiple reference voltages simultaneously.
- Generates a **thermometer code** which is then converted to binary.
- Used in very high-speed systems: oscilloscopes, RF front-ends, high-frequency data acquisition.

Working Principle

- A resistor ladder creates evenly spaced reference voltages.
- Each comparator checks if:

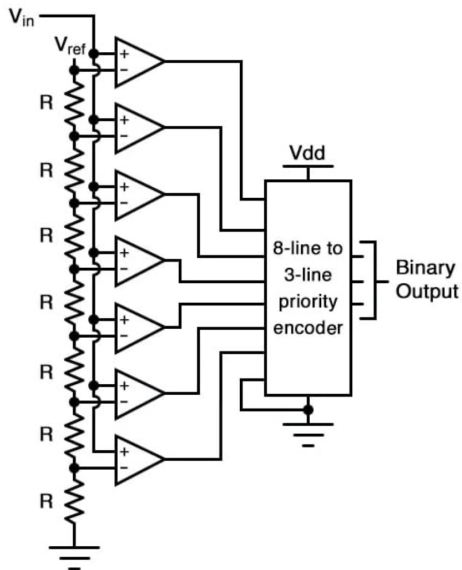
$$V_{in} > V_{ref,k}$$

- When input increases, more comparator outputs become HIGH $\rightarrow$ forming a thermometer pattern:

1111110000

- The encoder converts this into the corresponding binary output.

Flash ADC Block Diagram



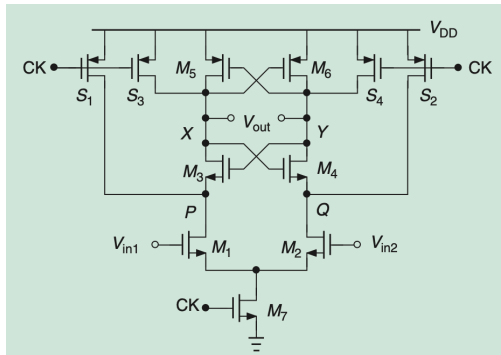
Resistor Ladder

- Provides **fixed reference steps** between V_{ref-} and V_{ref+} .
- For an N-bit Flash ADC: requires $2^N - 1$ reference levels.
- Example for 3-bit ADC $\rightarrow$ 7 comparators.
- Ladder output determines the threshold at which each comparator switches.

Why a StrongARM Latch Comparator?

- Fully dynamic → **no static power consumption.**
- Very high speed due to positive feedback regeneration.
- Simple topology, ideal for large comparator arrays in Flash ADC.

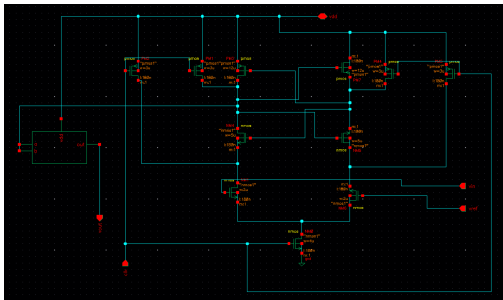
StrongARM Comparator Circuit



StrongARM Comparator Operation

1. Reset Phase ($\text{CLK} = 0$)
 - Output nodes are precharged.
 - No comparison occurs.
 - Comparator is prepared for the next evaluation.
2. Evaluation Phase ($\text{CLK} = 1$)
 - Input differential pair senses whether $V_{in} > V_{ref}$.
 - Positive feedback drives output to either logic HIGH or LOW.
 - Final output is a digital 1 or 0.

MOSFET Sizing Used in StrongARM Comparator



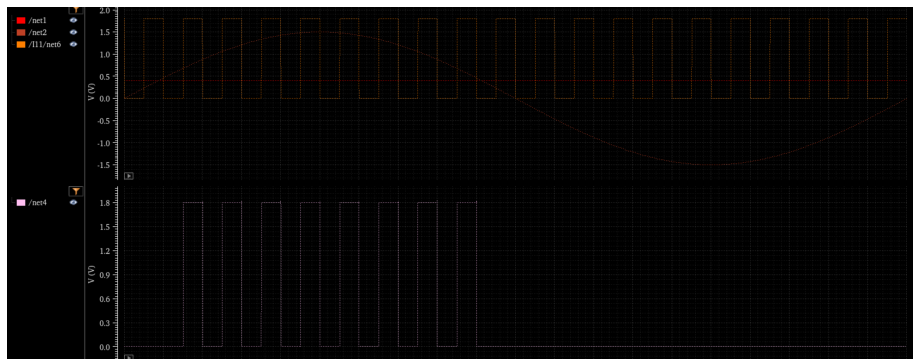
Transistors	Function	W (μm)	L (μm)
M1, M2	Input NMOS pair	2.0	0.18
M3, M4	Regenerative NMOS (latch)	5.0	0.18
M5, M6	Regenerative PMOS (cross-coupled)	12.0	0.18
M7	Tail / clock NMOS	4.0	0.18
S1-S4	Clock / pass transistors	3.0	0.18

Why These Sizes Are Used?

- **M1, M2 (Input Pair):** Kept moderately small to reduce input capacitance and avoid loading the resistor ladder. Enough width for sensing small voltage differences.
- **M3, M4 (Regenerative NMOS):** Larger width provides stronger positive feedback during evaluation, speeding up decision time.
- **M5, M6 (Regenerative PMOS):** Sized even wider because PMOS mobility is lower; this ensures fast pull-up and strong regeneration.
- **M7 (Tail NMOS):** Provides evaluation current. Sized larger than M1/M2 so the latch regenerates quickly but not too large to cause kickback noise.
- **Clock / Pass Transistors (S1–S4):** Medium sizing ensures low voltage drop during precharge and fast reset, without adding unnecessary load.

output of comparator

- net1,net2 are inputs
- net3 CLK
- net4 output



Thermometer Code

- Comparator array output looks like:

11111000

- The transition point identifies the digital output.
- Advantages: easy to interpret, monotonic.
- by using priority encoder made by cmos logic we convert it into binary
- because priority encoder takes the highest active '1' in the thermometer code.
- Converts it to binary using CMOS logic gates:

11111000 $\rightarrow$ 101

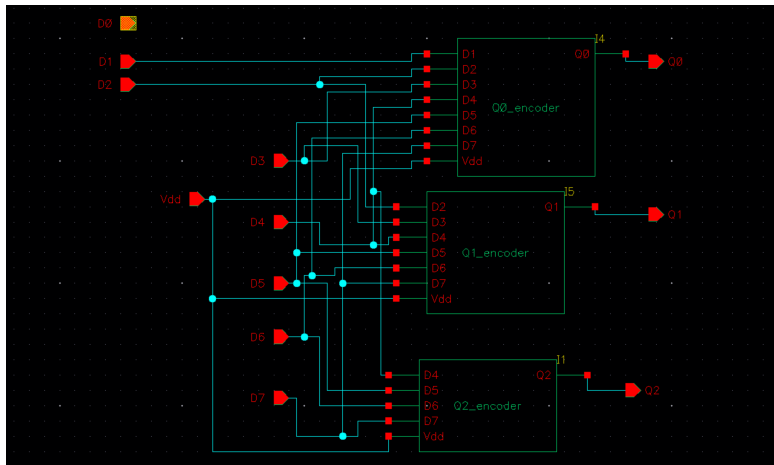
- Logic uses OR, AND, and NOT gates.

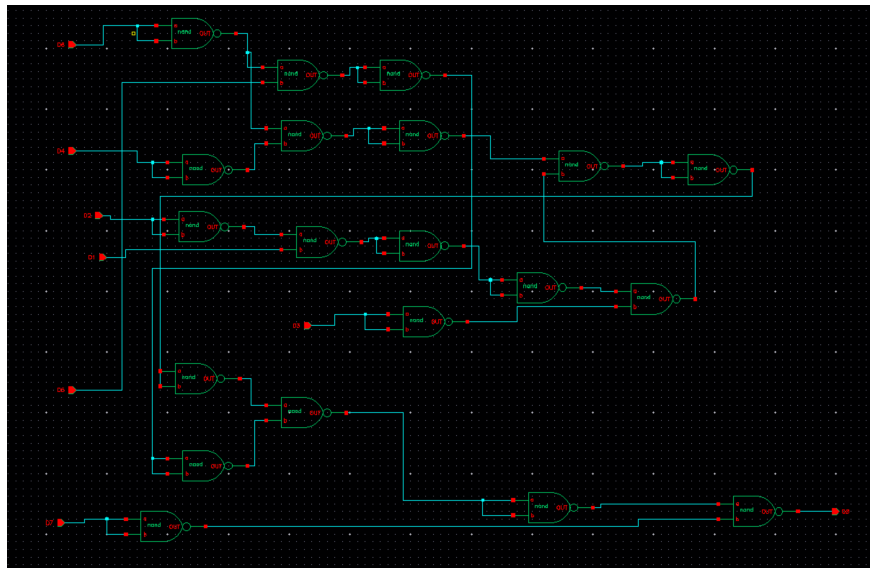
Truth Table: Thermometer Code to Binary Output

Thermometer Code (C7 C6 C5 C4 C3 C2 C1)	Decimal	Binary (B2 B1 B0)
0000000	0	000
1000000	1	001
1100000	2	010
1110000	3	011
1111000	4	100
1111100	5	101
1111110	6	110
1111111	7	111

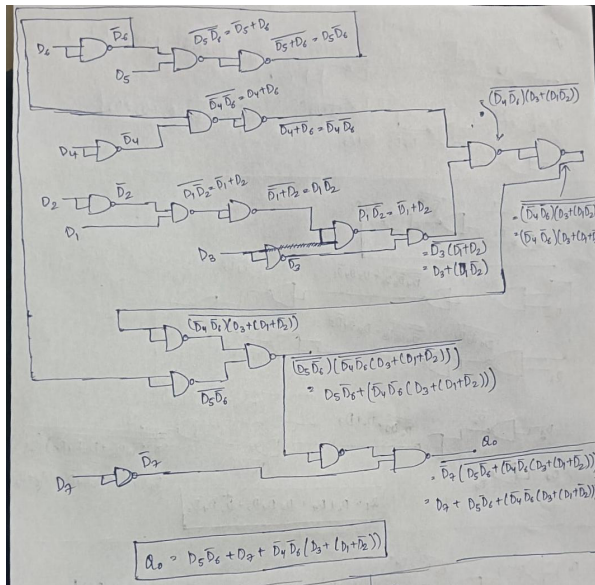
- This is the final digital output of the Flash ADC.

encoder circuit

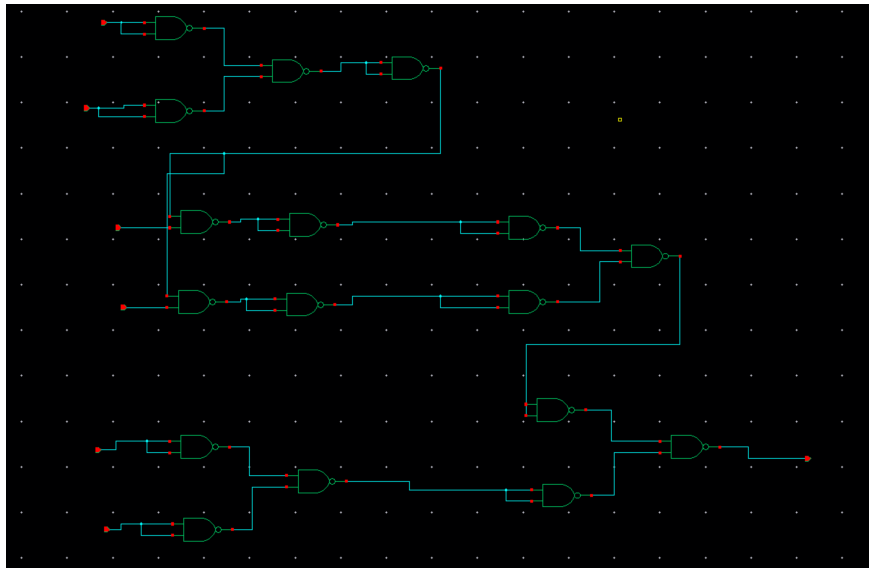




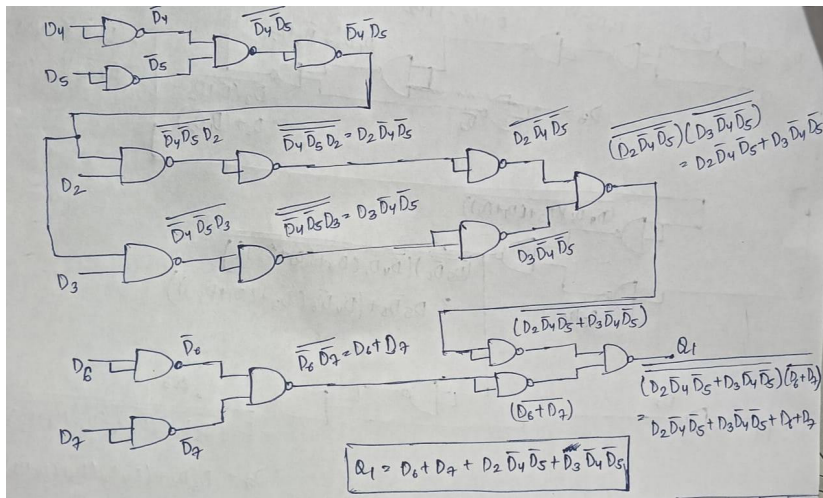
Q0

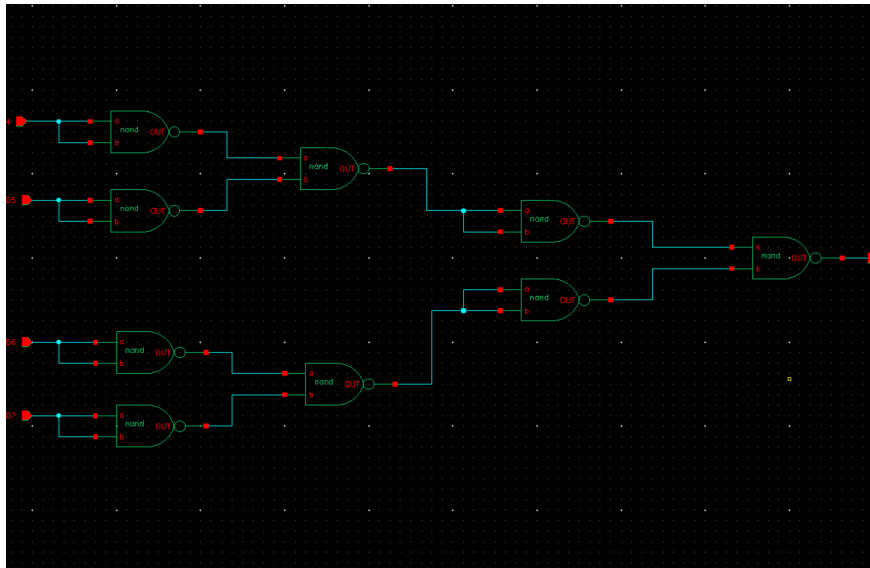


Q_1

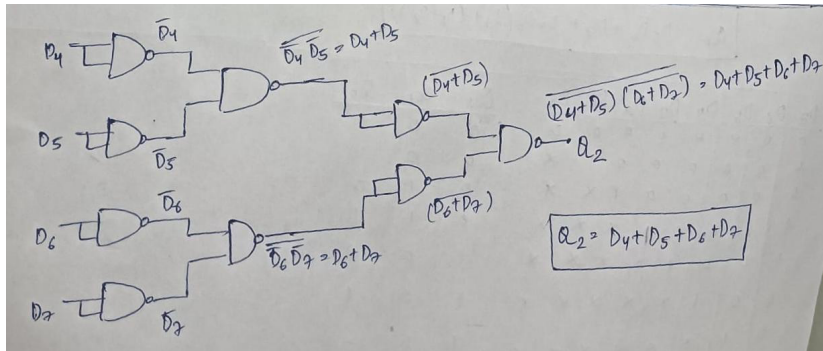


Q1





Q2



CMOS Logic Gate Sizing

Device	W (μm)	L (μm)	W/L
Logic NMOS (typical)	0.54	0.18	3.00
Logic PMOS (typical)	1.08	0.18	6.00

- These sizes (540 nm NMOS, 1080 nm PMOS) give a **PMOS 2 $\times$ NMOS** width ratio to compensate lower hole mobility and achieve balanced rise/fall drive.
- Using $L = 0.18 \mu\text{m}$ yields simple W/L ratios (3 and 6) convenient for schematic-level CMOS logic simulation.
- Good for encoder/combinational gates where driving speed and balanced transitions matter more than absolute matching.

Why Only 2-Input NAND Gates Were Used

2-input NAND gates were chosen for the encoder because:

- larger gates (3-input or 4-input) create long transistor stacks, which increases delay,
- stacked NMOS/PMOS devices cannot drive full logic levels and may clip the output (e.g., stopping at $\approx 1.0\text{ V}$ instead of 1.8 V),
- they load the comparator outputs heavily and disturb the StrongARM regeneration,
- and 2-input gates give faster switching, better noise margins, and easier sizing.

Thus, using only 2-input NAND gates ensures reliable logic levels and prevents glitches in the Flash ADC.

Resistor Ladder Voltages

- $V_{ref+} = 1.8 \text{ V}$, $V_{ref-} = 0 \text{ V}$
- All resistors: $R = 100 \text{ } \Omega$
- Step voltage:

$$V_{\text{step}} = \frac{1.8 \text{ V}}{8} = 0.225 \text{ V}$$

- Ladder node voltages:

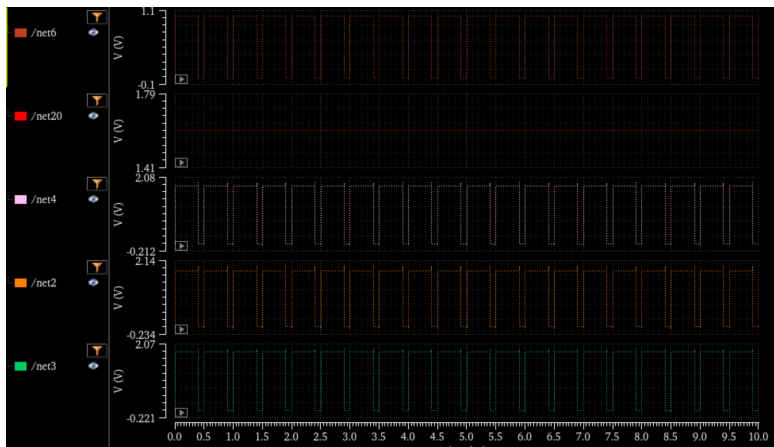
Node	Voltage (V)
V_1	0.225
V_2	0.450
V_3	0.675
V_4	0.900
V_5	1.125
V_6	1.350
V_7	1.575

Flash ADC Output

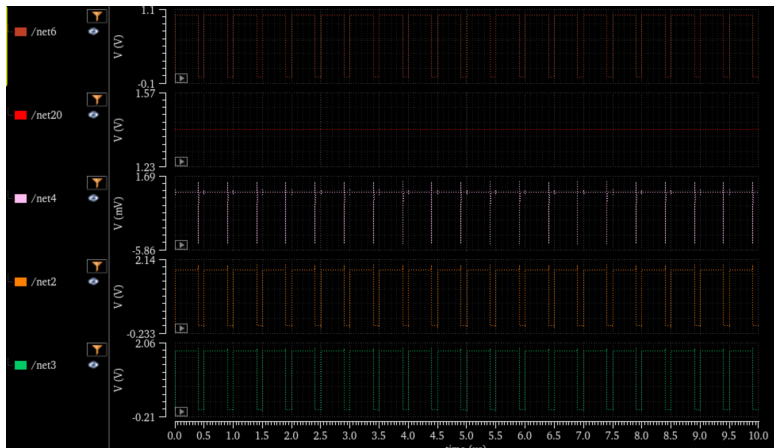
Input Range (V)	Thermometer	Binary
0–0.225	0000000	000
0.225–0.450	1000000	001
0.450–0.675	1100000	010
0.675–0.900	1110000	011
0.900–1.125	1111000	100
1.125–1.350	1111100	101
1.350–1.575	1111110	110
1.575–1.800	1111111	111

output at $v_{in} = 1.6V$

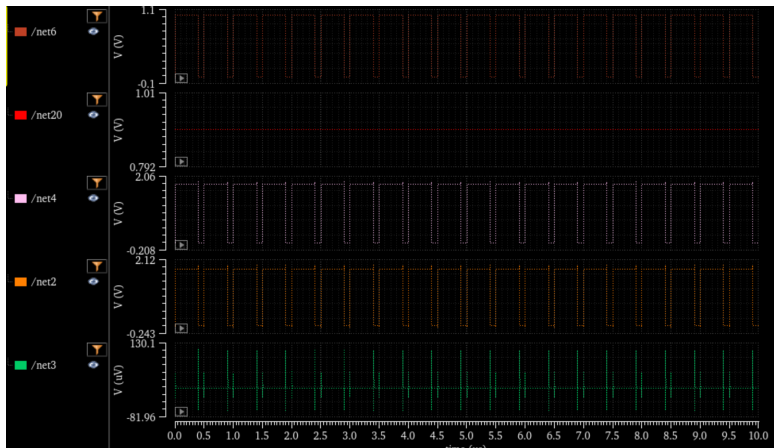
- net6 = CLK
- net20 = V_{in}
- net4 = Q_0 , net2 = Q_1 , net3 = Q_2



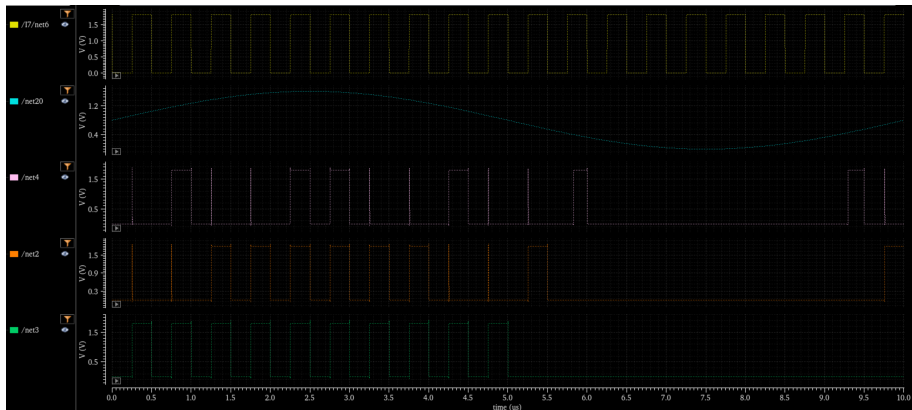
output at $v_{in} = 1.4V$



output at $v_{in} = 0.9V$



sin wave as input $v_{dc}=0.8\text{v}$ and amplitude= 0.8v



Output Spikes During Transitions

Small spikes or glitches may appear at the Flash ADC output during code transitions because:

- comparators resolve at slightly different times due to mismatch and regeneration delay,
- the thermometer-to-binary encoder momentarily sees invalid intermediate codes,
- and dynamic nodes in the StrongARM latch experience kickback noise during evaluation.

These effects cause short, harmless glitches that disappear once all comparator outputs settle.

Conclusion

we implemented a complete Flash ADC, consisting of:

- a resistor ladder generating seven reference levels from a 1.8 V input range,
- a StrongARM dynamic latch comparator array for high-speed parallel comparison,
- and a CMOS logic priority encoder that converts the thermometer code into a 3-bit binary output. only by using nand logic

The design achieves fast, low-power conversion by using fully dynamic comparators and compact CMOS logic.

References

- https://www.seas.ucla.edu/brweb/papers/Journals/BR_Magzine4.pdf.
- <https://www.allaboutcircuits.com/textbook/digital/chpt-13/flash-adc/>